

Introduction

Pengyang Shao received his Ph.D. degree in July 2025 from Hefei University of Technology under the supervision of Prof. Meng Wang. As a Ph.D. candidate, his research interests were related to information retrieval, including recommender systems and cognitive diagnosis. He is currently a visiting scholar at NExT++ Lab, NUS, under the supervision of Prof. Tat Seng Chua, focusing on LLM unlearning/safety.

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Research Summary

- 1) **LLM Unlearning**: Developing better LLM unlearning, including (i) a DRO-theoretic framework for balanced LLM unlearning (WWW 26) and (ii) a principled analysis of current forget losses for knowledge unlearning beyond input–output disassociation.
- 2) **Fairness & Privacy for Recommender Systems (RS)**: Studying statistical fairness based on adversarial learning; counterfactual user-side fairness based on potential outcome frameworks, and privacy attacks that make user preferences unlearnable.
- 3) **Cognitive Diagnosis (CD)**: Improving graph-based CD via information bottleneck principles and designing LLM based agents for BFS-driven concept augmentation to enhance CD under sparse concept descriptions.

First-Author & Corresponding-Author Publications

*** denotes the corresponding author. For your convenience, I have attached the proofs and texts of papers that are not yet searchable online, including the KDD 2026 paper and a collection of manuscripts currently under review.**

Accepted:

- **Pengyang Shao**, Lei Chen, et al. Multi-Agent Debate based Concept Augmentation for Enhanced Cognitive Diagnosis. KDD 2026 August Cycle, just accepted.
- [Jilong Liu, Pengyang Shao*](#), et al. [Debate over Mixed-knowledge: A Robust Multi-Agent Framework for Incomplete Knowledge Graph Question Answering](#). AAAI 2026.
- [Pengyang Shao, Yonghui Yang, et al. Exploring Heterogeneity and Uncertainty for Graph-based Cognitive Diagnosis Models in Intelligent Education](#). KDD 2025.
- [Pengyang Shao, Le Wu, et al. Average User-side Counterfactual Fairness for Collaborative Filtering](#). ACM TOIS, 2024.
- [Pengyang Shao, Le Wu, et al. FairCF: Fairness-aware collaborative filtering](#). Science China Information Sciences (SCIS), 2022.
- [Pengyang Shao, Zihan Wang, et al. Towards Reliable Cross-Domain Recommendation: A Disentangled Global Graph Learning based Framework](#). Frontiers of Computer Science (FCS), 2025.

- [Pengyang Shao, Chen Gao, et al. Breaking student-concept sparsity barrier for cognitive diagnosis. Frontiers of Computer Science \(FCS\), 2025.](#)

Under Review:

- **Pengyang Shao**, Le Wu, et al. Privacy Matters: Data Attack for Unlearnable User Preference in Recommendation. Under review at ACM TOIS. (Minor revisions)
- **Pengyang Shao**, Naixin Zhai, et al. BalDRO: A Distributionally Robust Optimization based Framework for Large Language Model Unlearning. Under review at WWW 2026.
- Chao Chen, **Pengyang Shao***, et al. Enhancing Cold-Start Personalization for Large Language Models with Social Information. Under review at ACL 2026.
- Naixin Zhai, **Pengyang Shao***, et al. Maximizing Local Entropy Where It Matters: Prefix-Aware Localized LLM Unlearning. Under review at ACL 2026.

Experiences

1) National University of Singapore Advisor:	Visiting Tat Seng Chua	2025.08-Present Singapore
2) Hefei University of Technology Advisor:	Ph.D. Meng Wang	2019.09 – 2025.06 Hefei, China
3) Tsinghua University - FIB Co-Advisor:	Intern Yong Li	2023.08 - 2024.03 Beijing, China

Academic Services

Journal/Conference Reviewer: IEEE TKDE, ACM ToRS/TOIS, IJCAI 2024, KDD 2024/2025/2026, WWW 2024/2026, AAAI 2026 et al.

Academic Mentorship

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- (1) Leading **Zhiyuan Han (USTC, PhD candidate)** on a GUI Agent research project in Beijing, providing core ideas, technical guidance.
 - (2) Mentoring **Jilong Liu (HFUT, PhD candidate)** toward an AAAI 2026 publication, contributing to problem formulation, method design, and paper writing.
 - (3) Mentoring **Chao Chen (HFUT, Master's student)** and **Naixin Zhai (USTC, Master's student)** to complete two ACL 2026 submissions, providing end-to-end guidance from methodology development to manuscript revision.

Awards

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- China Association for Science and Technology (CAST) Young Talent Support Program (PhD Special Track) 2024
 - First-Class Graduate Academic Scholarship (awarded annually) 2021-2024
 - Outstanding Graduate of Anhui Province 2019